

Wildfire Analysis

Chernobyl Exclusion Zone (Red Forest) – April 2020



The Event:

There were a series of wildfires that occurred within the Chernobyl Exclusion Zone during April 2020 [1]. These wildfires are of significant importance because this is an area of high radioactivity in and above ground, in which the radioactive particles are released into the air during wildfire events. This can not only increase the amount of dangerous particles in the air for those who live near the areas affected by the wildfires but also 100's of km's away where the smoke particle clouds travel [2]. This also increases the danger to the hundreds of firefighters who are forced to go in this exclusion zone to mitigate and extinguish these fires. Although the Chernobyl Exclusion Zone wildfires were extensive, I will focus more on the area near the Chernobyl reactor and Red Forest area which has higher ground radioactivity compared to the rest of the region.

The fires began in early April and were reported to last sometime after the end of April. From Sentinel-3 imagery it can be seen that the major fires in the Red Forest area near Chernobyl started between acquisition of the imagery from April 8th and April 9th.

It was reported that heavy rains on April 14th greatly helped put out a majority of the fires



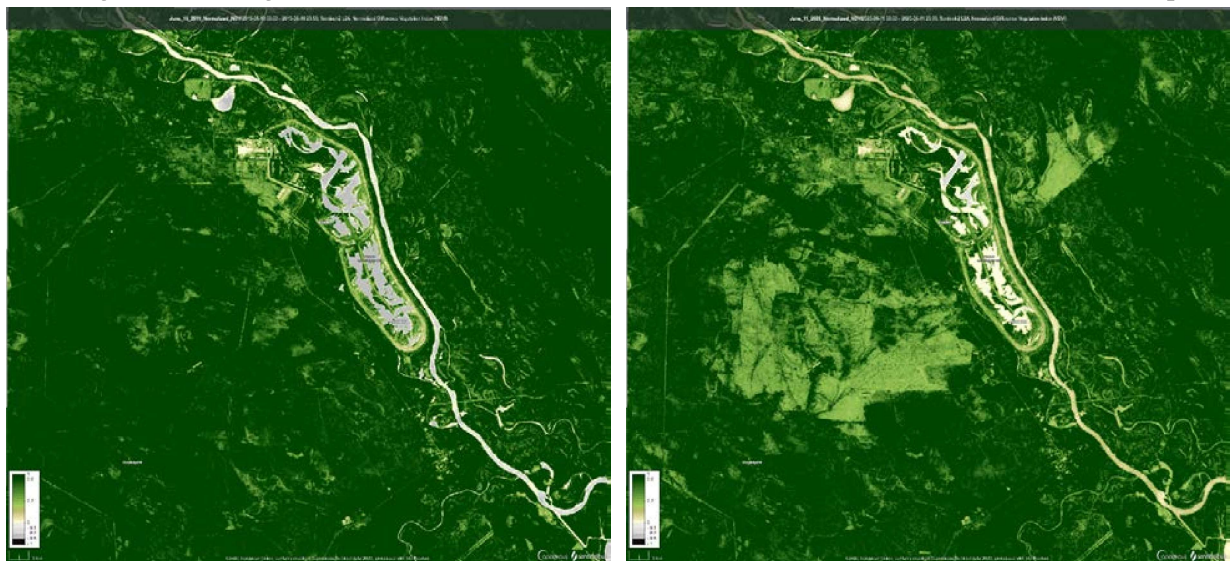
that were raging in the exclusion zone [3]. This is further evidenced by the lack of usable satellite data to observe the ongoing changes of the fires (due to high or complete cloud cover).

It is unclear when the wildfires officially ceased to be as little news is available after the initial reports that the fires closer to the Chernobyl power plant were put out. Early news reported it to be contained after the heavy rains on April 14th but according to several sources there were still fires going on in the Chernobyl Exclusion zone at least until the end of April as reported by the large amount of smog in the city of Kyiv from the fires on April 26th [4] [5]. The most visible (though still moderately cloudy) Sentinel-2 image after the initial wildfires occurs on April 22nd, but afterwards every acquisition for the rest of April and May is not very visible due to heavy cloud cover and it is not until early June that the area is visible again through Sentinel-2 imagery making it difficult to determine the actual end of the fires.

The Aftermath

Luckily these fires never reached the Chernobyl Reactor 4 area. And although there was an increase of radiation particles in the air, it was not deemed high enough to warrant worry according to the International Atomic Energy Agency [6]. If the

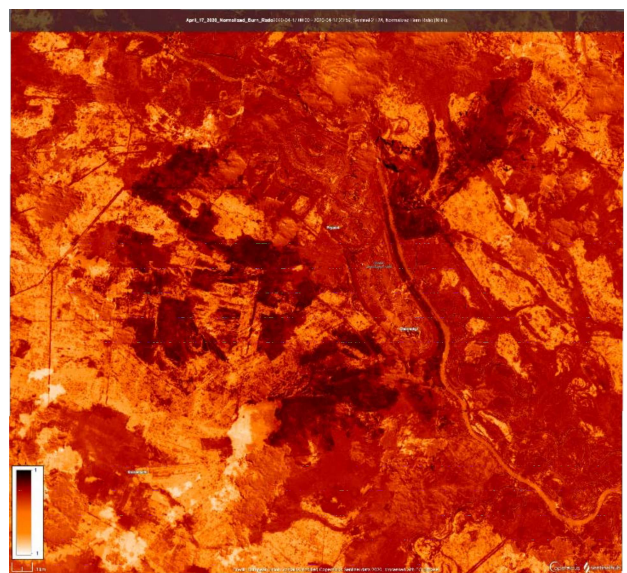
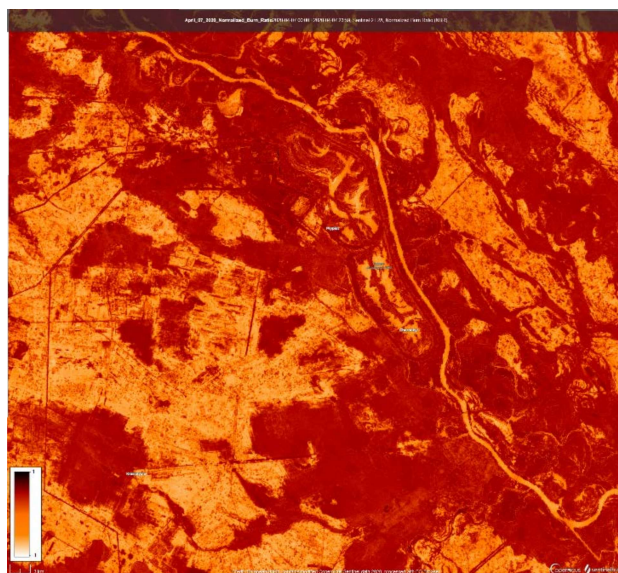
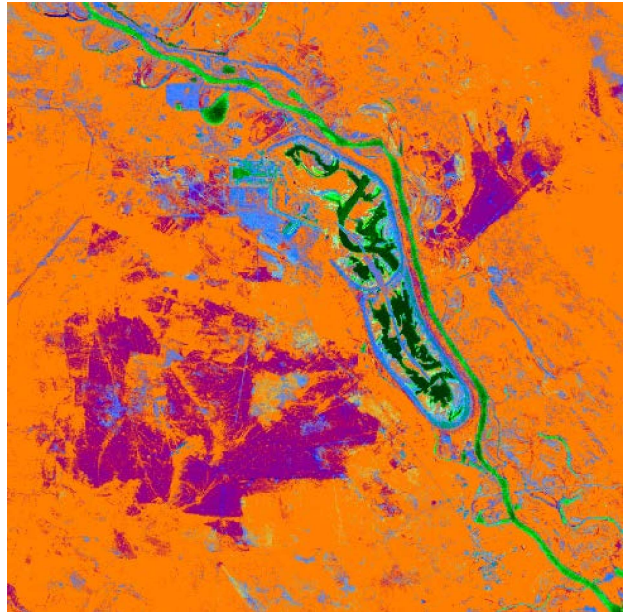
Looking at the change in Enhanced NDVI between June 10, 2019 and June 11, 2020 it is quite



clear to see the level of vegetation that was lost by the wildfires in April 2020 in just the area near Chernobyl. Especially those areas in the Red Forest zone south of Prypiat and Chernobyl Reactor 4.

With a change detection analysis of this vegetation it becomes more vivid the areas that were impacted by the fires in this region. In the image to the left, the Dark Red/Magenta areas being the areas with the greatest change after June 10, 2019 to a year later around the same seasonality of June 11, 2020.

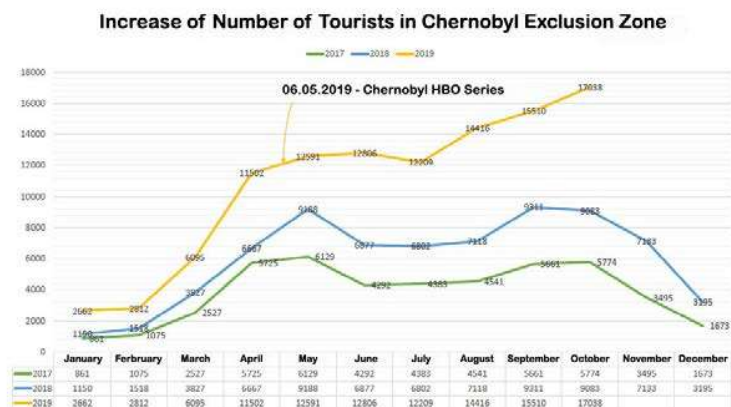
This is further represented by looking at the imagery from the Normalized Burn Ratio and imagery from Sentinel-2. As seen below, the burned areas have a low reflectance in the NIR and SWIR wavelengths thus looking like a deep dark red in the latter image.



The wildfires not only have an impact on the ecology and health of the people living in Ukraine and near this exclusion zone, but it was reported that 30% of the tourist attractions were destroyed in the wildfires in April 2020, most notably an old Soviet children's summer camp with vintage and picturesque imagery of Soviet-era iconography [7] [8]. It is additionally unclear if more of these attractions in the

region were destroyed after this initial report from the first wildfires.

Tourism to and within the Chernobyl Exclusion Zone was over 100,000 tourists as of 2019 [9]. A number that has been increasing greatly year after year since it's opening for tourism in 2010 [10]. Reportedly nearly 80% of these tourists come from outside of the Ukraine. The cost of going on a tour within this zone ranges from 100 to 250 USD for non-Ukrainian tourists for a 1 day trip. This revenue for Ukrainian tourism would account for between 8 to 20 million USD just on this day trip alone, not accounting for the other revenue brought to Ukraine in the form of accommodations, food, travel, etc. With the wildfires destruction of 30% of the tourist attractions, this could lead to a large decrease in the potential revenue of the area. For a country with a GDP that accounts for only 0.13% of the global economy and the 34th biggest country in the world, this loss in economic tourism could impact this region of Ukraine greatly [11].



Additional Analysis or other Limitations

- Unable to acquire the Chernobyl Exclusion Zone nor Red Forest area datasets i.e. vector polygons of these areas
 - o Zonal statistics analysis to get a better assessment of the extent and percentage of land burned from the fire damage within the Chernobyl Exclusion Zone (and additionally Red Forest region within the zone).
 - o Would have allowed for determining the percentage of red forest area burned to calculate areas of higher radioactivity as opposed to those burned with lower levels of radioactivity.
 - o Would allow for better visualization in the areas of interest
- Lack of data of geo-positional vector information (Chernobyl Reactor, town of Pripjat, etc.) which would have been great to have for reference on the Satellite images (other than the nametags)
 - o This would drive the point across better the proximity of the wildfires to these locations.
- Sentinel Hub labelling for locations is a bit off when utilizing saving area of interests as images
- Daily Timelapse videos from Sentinal Hub for various factors (Carbon Monoxide imagery from Sentinel-5, Sentinel-3 OLCI) give a good idea of the extent and impact over time during the event

- A larger report and more time would allow for more information, such as getting a rough estimate of the radiation potential released from the ground if the wildfires went on for longer or with higher intensity and/or had reached closer to Chernobyl reactor 4

Sources:

- [1] <https://www.euronews.com/2020/04/10/village-evacuated-as-forest-fires-in-chernobyl-exclusion-zone-continue-to-burn>
- [2] <https://www.sckcen.be/en/news/forest-fires-chernobyl-increased-radioactivity-near-fires-no-danger-population>
- [3] <https://abcnews.go.com/International/ukraine-wildfires-close-chernobyl-extinguished-rain-falls/story?id=70138987>
- [4] <https://www.bbc.com/news/world-europe-52329498>
- [5] <https://www.dw.com/en/chernobyl-fires-still-burning-on-anniversary-of-accident/a-53253968>
- [6] <https://www.reuters.com/article/us-ukraine-chernobyl-fire-iaea/fires-near-chernobyl-pose-no-risk-to-human-health-iaea-says-idUSKCN2262YU>
- [7] <https://www.forbes.com/sites/katyagorchinskaya/2020/04/15/fire-destroys-a-third-of-tourist-attractions-in-chernobyl/#58b43fed2467>
- [8] <https://www.chernobyl.one/emerald-summer-camp/>
- [9] <https://ukraine-kiev-tour.com/2019/chernobyl-tours-statistics.html>
- [10] Banaszkiewicz, Magdalena & Kruczek, Zygmunt & Duda, Anna. (2017). THE CHERNOBYL EXCLUSION ZONE AS A TOURIST ATTRACTION. REFLECTIONS ON THE TURISTIFICATION OF THE ZONE. Folia Turistica. 44. 145-169. 10.5604/01.3001.0010.8736
- [11] <https://tradingeconomics.com/ukraine/gdp>